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1. Electromagnetic Spectrum & Wavelength Order

- Sabse lamba wavelength → Radio Waves; sabse chhota wavelength → Gamma Rays. [Q1426]
- Wavelength decreasing order: Radio > Microwave > Infrared > Visible > Ultraviolet > X-Rays > Gamma. [Q1426]
- Gamma rays sabse chhota wavelength aur sabse zyada energy rakhti hain. [Q1426]
- Wavelength ka Greek letter Lambda (λ) hai; visible light ka range 400 nm se 700 nm tak hota hai. [Q1427]
- X-rays ki khoj Wilhelm Conrad Roentgen ne ki; gamma rays ki khoj Paul Villard ne; infrared ki khoj Sir Frederick William Herschel ne. [Q1426]
- X-rays ka use haddi mein fracture check karne ke liye hota hai. [Q1431]
- Electromagnetic waves ke examples: radio, microwave, infrared, visible, UV, X-rays, gamma. [Q1430, Q1431]

2. Wave Terminology & Definitions

- Wavelength (λ) = do kramagat crests ya do kramagat troughs ke beech ki doori. [Q1427, Q1440, Q1450]
- Frequency (f) = prati second oscillations ki sankhya; unit Hertz (Hz). [Q1437, Q1446]
- Amplitude = mean position se maximum displacement; crest aur trough donon opposite directions mein maximum displacement hain. [Q1448]
- Time period = ek poora oscillation complete karne mein laga samay. [Q1437]
- Pulse = short duration ki wave; non-sinusoidal waveform jaise square wave. [Q1445]
- Wavelength = do kramagat compressions ya do kramagat rarefactions ke beech ki doori (longitudinal wave mein). [Q1450]

3. Transverse vs Longitudinal Waves

- Transverse waves mein particles perpendicular direction mein vibrate karte hain wave propagation ke. [Q1428, Q1429, Q1449]

- Transverse examples: water surface ripples, seismic S-waves, electromagnetic waves (radio, light). [Q1428, Q1429]
- Longitudinal waves mein particles parallel direction mein vibrate karte hain wave propagation ke. [Q1434, Q1438, Q1450]
- Longitudinal examples: sound waves, ultrasonic waves, seismic P-waves. [Q1434, Q1438]
- Seismic S-waves (shear waves) = transverse displacement; P-waves = compressional/longitudinal. [Q1429]
- Rayleigh waves = surface waves jo solids ki surface ke paas travel karti hain. [Q1429]

4. Sound Waves: Properties & Medium

- Sound waves mechanical waves hain — inhe travel karne ke liye medium chahiye; vacuum mein travel nahi kar sakti. [Q1436, Q1444]
- Sound ki speed order: Solids > Liquids > Gases; vacuum mein zero. [Q1444]
- Sound ke liye particle vibration aur energy transfer zaroori hai; medium ke bina disturbance transfer nahi ho sakta. [Q1436]
- Jab sound ek medium se doosre mein jaati hai (e.g., air to water), uski frequency same rehti hai lekin speed aur wavelength badalte hain. [Q1447]
- Formula: $v = f \times \lambda$, jahan v = speed, f = frequency, λ = wavelength. [Q1442, Q1447, Q1450]

5. Ultrasound: Properties, Applications & Echolocation

- Ultrasound ki frequency >20 kHz hoti hai — human hearing limit se upar. [Q1430, Q1432, Q1439]
- Ultrasound mechanical wave hai, electromagnetic nahi. [Q1430]
- Ultrasound obstacles ki presence mein bhi well-defined paths mein travel kar sakti hai. [Q1432, Q1439]
- Ultrasound se defects detect karte hain: transmitter wave bhejta hai, detector receive karta hai; received wave mein kami = fault. [Q1432]
- Industrial applications: drilling holes, metal defect detection, ultrasonic cleaning. [Q1439]
- Ultrasound produce karte hain: dolphins, bats, porpoises. [Q1439]
- Bats echolocation use karte hain: high-frequency ultrasonic squeaks emit karte hain, obstacle se reflect hokar echo sunke navigate karte hain. [Q1435]

6. Infrared & UV Radiation

- Infrared (IR) = heat energy; har object 0 K (−273.16°C) se upar temperature par emit karta hai. [Q1441]
- IR wavelength range: 710 nm se 1 mm tak. [Q1441]
- Garam furnace se emit hone wali radiation infrared hoti hai. [Q1441]
- UV radiation = Sun se emit hone wali harmful radiation; wavelength 400 nm se 0.6 nm tak. [Q1443]
- UV ko special lamps aur very hot bodies produce karte hain. [Q1443]
- UV ki overexposure se melanin production badhta hai aur skin tan hoti hai. [Q1443]
- Pathology labs mein test tubes sterilize karne ke liye UV radiation use hoti hai bacteria eliminate karne ke liye. [Q1433]
- Microwave wavelength range: 0.1 m se 1 mm tak. [Q1433]

7. Wave Relationships ($v = f\lambda$)

- Same medium mein speed constant rehti hai; frequency aur wavelength inversely proportional hain. [Q1442]
- Frequency double karne par wavelength half ho jaati hai. [Q1442]
- Sound ek medium se doosre mein jaaye to frequency unchanged rehti hai, speed aur wavelength adjust hote hain. [Q1447]
- Mathematical relation: $\lambda = v/f$. [Q1440, Q1450]

8. ⚠ Traps / Key Differences Summary (MCQ Q1426–Q1450)

- ⚠ Trap: Gamma rays sabse chhota wavelength, radio waves sabse lamba — confuse mat hona. [Q1426]
- ⚠ Trap: Ultrasound mechanical wave hai — electromagnetic wave nahi; isliye EM spectrum mein nahi aati. [Q1430]
- ⚠ Trap: Alpha aur beta rays radioactive particles hain, electromagnetic waves nahi. [Q1431]
- ⚠ Trap: Bats ultrasonic (>20 kHz) use karte hain, infrasonic nahi. [Q1435]
- ⚠ Trap: Sound longitudinal hai, light transverse hai — dono ka nature alag hai. [Q1428, Q1434]
- ⚠ Trap: Frequency medium change hone par same rehti hai, speed aur wavelength badalte hain. [Q1447]
- ⚠ Trap: Hertz unit hai (Hz), characteristic nahi — exam mein confuse karne ke liye dete hain. [Q1446]
- ⚠ Trap: Pesticide removal ultrasound ka industrial application nahi hai. [Q1439]
- ⚠ Trap: Garam furnace se infrared emit hoti hai, X-ray ya microwave nahi. [Q1441]
- ⚠ Trap: Seismic S-waves = transverse; P-waves = longitudinal — S ka matlab shear (transverse), P ka matlab primary (longitudinal). [Q1429]

9. Numerical: Frequency from Vibrations / Crests & Troughs

- Frequency (f) = Number of vibrations / Time taken; unit Hz. [Q1451, Q1452, Q1453, Q1454]
- 40 vibrations in 2.5 s $\rightarrow f = 40/2.5 = 16$ Hz. [Q1451]
- 40 crests + 40 troughs in 0.3 s $\rightarrow 40$ complete cycles $\rightarrow f = 40/0.3 = 133.33$ Hz. [Q1452]
- 20 crests + 20 troughs in 0.4 s $\rightarrow 20$ complete cycles $\rightarrow f = 20/0.4 = 50$ Hz. [Q1453]
- 10 crests + 10 troughs in 0.1 s $\rightarrow 10$ complete cycles $\rightarrow f = 10/0.1 = 100$ Hz. [Q1454]
- Yaad rakhna: 1 crest + 1 trough = 1 complete cycle; number of crests = number of cycles. [Q1452, Q1453, Q1454]

10. Numerical: Wave Equation $v = f\lambda$

- Formula: Velocity (v) = Frequency (f) $\times$ Wavelength (λ); isse $\lambda = v/f$ aur $f = v/\lambda$ nikaala ja sakta hai. [Q1455, Q1457, Q1458, Q1459, Q1460]
- Distance 10 m in 0.04 s, $f = 200$ Hz $\rightarrow v = 10/0.04 = 250$ m/s; $\lambda = 250/200 = 1.25$ m. [Q1455]
- $f = 71 \times 10^4$ Hz, $v = 3 \times 10^8$ m/s $\rightarrow \lambda = (3 \times 10^8)/(71 \times 10^4) = 30000/71 = 422.5$ m. [Q1457]
- Crests 80 m apart ($\lambda = 80$ m), $v = 20$ m/s $\rightarrow$ Time period $T = \lambda/v = 80/20 = 4$ s. [Q1458]
- $\lambda = 100$ m, $v = 25$ m/s $\rightarrow f = v/\lambda = 25/100 = 0.25$ Hz. [Q1459]
- $\lambda = 125$ m, $v = 25$ m/s $\rightarrow f = v/\lambda = 25/125 = 0.20$ Hz. [Q1460]

11. Numerical: Frequency & Time Period Relation

- Time period (T) = $1/\text{frequency (f)}$; yaani $T = 1/f$ aur $f = 1/T$. [Q1456]
- Ye inverse relation hai — jitni zyada frequency utna kam time period. [Q1456]

12. Numerical: Formula Cheat Sheet

- Frequency: $f = \text{Number of vibrations} / \text{Time}$ — unit Hz.
- Time Period: $T = 1/f$ — unit second (s).
- Wave Velocity: $v = f \times \lambda = \text{distance/time}$ — unit m/s.
- Wavelength: $\lambda = v/f$ — unit meter (m).
- Frequency (alternate): $f = v/\lambda$ — unit Hz.

13. ⚠ Traps / Key Differences Summary (Numerical Q1451–Q1460)

- ⚠ Trap: Crests aur troughs count karne par 1 crest + 1 trough = 1 complete cycle hota hai; sirf crests ki sankhya = cycles ki sankhya. [Q1452, Q1453, Q1454]
- ⚠ Trap: Distance/time se velocity nikalti hai, frequency nahi; velocity ke baad $\lambda = v/f$ se wavelength nikaalni padti hai. [Q1455]
- ⚠ Trap: $\lambda = 80$ m aur $v = 20$ m/s se time period (T) nikalti hai (4 s), frequency nahi; $f = 1/T = 0.25$ Hz alag calculation hai. [Q1458]
- ⚠ Trap: $\lambda = 100$ m, $v = 25$ m/s $\rightarrow f = 25/100 = 0.25$ Hz, 25 Hz nahi; divide karne par value 1 se kam aa sakti hai. [Q1459]
- ⚠ Trap: Broadcasting station question mein scientific notation handle karna: 71×10^4 ko direct divide karo 3×10^8 se; 10^4 upar jaake $10^{8-4} = 10^4$ ban jaata hai. [Q1457]